



Understanding Major Topics and Attitudes Toward Deepfakes: An Analysis of News Articles

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Abstract. Deepfakes, highly realistic fake videos or images created using deep learning algorithms, have raised concerns due to their potential for malicious use and the dissemination of false information. This paper aims to analyze news articles related to deepfakes, focusing on the major topics discussed and the attitudes expressed toward this emerging technology. Understanding the public's awareness, concerns, and reactions toward deepfakes is crucial in shaping informed responses. This study conducted exploratory data analysis, topic modeling analysis, and sentiment analysis by examining 4,920 news articles from the Nexis Uni database between 2000 and 2022. The topics discussed encompass various domains, including politics, business, entertainment, and more. We uncovered how deepfakes can be used for manipulation, the risks they pose to industries, and their potential impact on public trust and society. Furthermore, sentiment analysis of the news articles allowed us to gauge the overall public perception of deepfakes. We examined the emotional tone and attitudes conveyed in the articles to determine whether deepfakes were portrayed as a threat, a tool for mischief, or a harmless form of entertainment. This analysis provides insights into the prevailing sentiments surrounding deepfakes and their potential implications. This study addresses the research gap on the evolution of deepfake topics and sentiments over time and across diverse contexts.

Keywords: Deepfakes · misinformation · news articles · topic analysis · sentiment analysis

1 Introduction

Deepfakes, a form of digital forgery, have recently gained significant attention due to their increasing prevalence. This phenomenon involves using deep learning algorithms to create remarkably realistic fake videos or images [1]. Individuals can manipulate visual content to make it appear like someone is saying or doing something they never actually did in real life by employing this technology. Although deepfakes can serve as a means of entertainment, there is a growing concern regarding their potential malicious use for purposes such as political sabotage, market manipulation, and the dissemination of false information [2, 3].

As the primary source of information for many individuals, online news plays a vital role in shaping attitudes and public perceptions [4, 5]. Notably, news articles discussing deepfakes have witnessed a significant rise in recent years. Consequently, analyzing news articles related to deepfakes is imperative as they reflect the public's awareness, concerns, and reactions toward this emerging technology [3]. We can gain valuable insights into the prevalence, frequency, and diverse range of deepfake-related issues and their potential implications for society and public opinion by examining news articles.

Analyzing news articles is a tool to gauge the current state of deepfake awareness among the general public [6]. News outlets often act as intermediaries between technology advancements and the wider society, offering insights, explanations, and opinions on various topics. News articles allow journalists and writers to convey information regarding deepfake technology's capabilities, potential risks, and societal impact. Therefore, understanding the major topics discussed within news articles allows us to identify the key concerns and areas of focus about deepfakes.

Moreover, news articles provide rich data on the public's attitudes and perceptions toward deepfakes [7]. Through analysis, we can gain an understanding of how the general public perceives and responds to this technology. Do they consider deepfakes a serious threat or a harmless tool for entertainment? Are they aware of the potential dangers associated with malicious use? By examining the tone, sentiment, and rhetoric employed in news articles, we can uncover the range of attitudes, concerns, and even public policy proposals that have emerged in response to deepfakes.

News articles also offer insights into the broader implications of deepfakes for society and public opinion. We can identify the emerging themes regarding deepfakes by examining the context in which deepfakes are discussed within news articles [8]. Political discourse, market dynamics, and public trust in media are a few examples of areas mentioned in news articles. Analyzing news articles allows us to explore the potential consequences and challenges that arise from the proliferation of deepfakes, aiding policymakers, researchers, and the public in formulating informed responses.

However, a research gap exists in understanding how deepfake topics and sentiments have evolved over time and across different sources and themes. While some studies have focused on specific aspects of deepfakes, such as detection methods [9] and social implications [10], to our knowledge, there is a lack of work that examines trends and patterns of deepfake news coverage over an extended period and across diverse contexts. This is important to address since it can help researchers and practitioners to understand the societal and historical factors influencing the public discourse and perceptions of deepfake technology. Therefore, this study aims to fill the above gap by analyzing 4920 new articles about deepfakes from the Nexis Uni database between 2000 and 2022.

The objectives of this paper are twofold. Firstly, we aim to identify the main topics and trends that emerge from news articles concerning deepfakes. We uncover the predominant themes, issues, and debates surrounding this technology by conducting a topic modeling analysis to understand the critical areas of focus within the public discourse on deepfakes. We also identify shifting narratives and emerging concerns or perspectives by tracking trends over time.

Secondly, we investigate the sentiments expressed in news articles related to deepfakes. Sentiment analysis allows us to explore the emotional tone and attitudes conveyed

within the articles, providing insights into the public's perception of deepfakes. Examining emotions - positive, negative or neutral, allows us to gain a nuanced understanding of how news outlets and journalists frame deepfakes. This analysis will shed light on whether deepfakes are predominantly portrayed as a threat, a tool for mischief, or a harmless form of entertainment.

2 Literature Review

2.1 Deepfakes Introduction and Their Societal Impacts

Deepfakes have emerged as a significant technological phenomenon with profound societal implications [11]. Deepfakes are artificial intelligence image synthesis technologies that produce videos that may imitate actual videos [12]. The term “deepfake” is derived from “deep learning” and “fake,” highlighting the role of deep learning algorithms in generating deceptive media. These fabricated media assets make it possible to manipulate visual content to convincingly portray individuals saying or doing things they never actually did in real life [13].

The societal impact of deepfakes stems from their potential to undermine truth, trust, and authenticity. One of the most concerning aspects of deepfakes is their ability to deceive viewers, blurring the line between real and fake [14]. The hyper-realistic nature of deepfakes makes it increasingly difficult for people to discern genuine content from manipulated or fabricated ones. This erosion of trust can have wide-ranging consequences, from damaging the credibility of individuals and public figures to fueling misinformation campaigns. Misinformation can violate one's right to privacy by harming that person's reputation under specific conditions [15], and by disregarding the privacy of others who are part of its target audience.

Political sabotage and manipulating public opinion are one of the most significant societal risks associated with deepfakes [16]. Deepfakes are misinformation campaigns that are weaponized to influence elections and cause unrest [3]. These campaigns create false or misleading videos of political leaders or candidates, making them appear to engage in compromising or unethical behavior. For instance, a deepfake video might depict a candidate accepting bribes, making inflammatory statements, or engaging in illicit activities [17]. The intention is to damage reputation, credibility, and public image, thereby influencing public perception and potentially swaying electoral outcomes. The potential impact of such malicious use of deepfakes on political stability and public trust in institutions is a cause for grave concern.

Deepfakes are often associated with negative implications and concerns about their potential misuse. However, it is essential to recognize that deepfakes also have the potential for positive societal impact in certain contexts. One of the positive applications of deepfakes lies in the realm of entertainment and creative expression. Deepfake technology allows for seamless face-swapping, enabling filmmakers and visual effects artists to bring characters to life in previously unimaginable ways [18]. This technology opens up possibilities for enhanced storytelling, enabling filmmakers to resurrect deceased actors or seamlessly blend real and fictional elements [19]. This creative application of deepfakes has the potential to captivate audiences and enhance the cinematic experience.

Deepfakes can also enhance learning experiences in educational settings [20]. Educators can create immersive simulations and interactive experiences that help people visualize historical events and enable art education using deepfake technology [21, 22]. For example, traditional methods of teaching history often rely on textbooks, images, or recorded audio to convey information about historical events and figures. However, deepfake technology can recreate historical figures or bring famous speeches to life, allowing individuals to engage with the material more vividly and memorably. These educational applications can potentially foster more profound understanding and engagement among learners.

2.2 News Articles and Deepfakes

News coverage of deepfakes has garnered significant attention as technology continues to evolve and impact various aspects of society. Journalists and news outlets are crucial in informing the public about the emergence, implications, and risks associated with deepfakes [21]. These articles often cover the latest advancements in deepfake technology, exploring the technical processes behind their creation, the challenges associated with detecting them, and the strategies to combat them [3]. These news articles have expressed worry that deepfakes would sabotage the democratic process, jeopardize national security, and continue the persistent problem of fake news and misinformation [7]. However, such news articles also help the general public understand the complexity of deepfakes and their potential consequences by presenting these technical aspects in accessible language.

News articles on deepfakes cover various topics, providing readers with different understandings of this emerging technology and its implications. These articles delve into multiple domains, including politics, business, entertainment, and more, to explore the multifaceted nature of deepfakes. In politics, news articles examine how deepfakes can be used for political manipulation and misinformation campaigns, threatening the basic information security of democratic polities [24]. They analyze the potential impact of deepfakes on elections, public perception of political leaders, and the erosion of trust in democratic processes [25]. These news articles contribute to the discussion on the vulnerability of political systems to deepfake threats and the need for safeguards and awareness among the public by highlighting these concerns.

In the business sector, news articles highlight the risks deepfakes pose to industries such as advertising, marketing, and brand reputation [26, 27]. They discuss the challenges media organizations face in verifying the authenticity of video content and the potential reputational risks associated with the spread of deepfakes. Entertainment is another area covered by news articles on deepfakes. While deepfakes can enhance storytelling and create stunning visual experiences, the potential for deepfakes to be utilized for misleading intentions is demonstrated in news articles, such as creating explicit or non-consensual content using people's likenesses without their permission [28]. These articles emphasize the need for businesses to stay informed and be proactive in addressing the risks posed by deepfakes, whether through technology-driven solutions, legal frameworks, or public awareness campaigns.

These news articles aim to raise individuals' awareness of the impacts and dangers of deepfakes in various contexts and explore the influence of deepfakes on individual credibility assessments of articles [6, 7]. Deepfake-related content and public service announcements like Jordan Peele's Obama deepfake aim to heighten viewer skepticism about political videos [29]. News articles also attempt to educate readers on deepfakes' potential for deception and how they might be used to mislead. Readers are encouraged to engage in scrutiny, actively assessing the credibility and potential manipulations present within media artifacts. These critical analyses extend beyond deepfakes, spurring a broader examination of media literacy and how information is consumed and shared [30].

3 Methodology

In this section, we elaborate on our dataset, followed by a brief description of text preprocessing. We then introduce exploratory data analysis, topic modeling analysis, and sentiment analysis as our research methods applied in this study.

3.1 Dataset and Text Preprocessing

We retrieved deepfake news articles from the Nexis Uni database using the search term "deepfake". We focused exclusively on newspapers with English content. The preliminary search resulted in 5,569 news articles published by 569 different sources, with dates ranging from April 27, 2000, to October 8, 2022. We extracted various data elements for analysis, including the news article's title, publication source, publication date, and content. Additionally, we gathered geographical information to gain insights into the relevant geographical regions associated with the content.

After eliminating duplicates, non-English content, and incomplete items, we narrowed our dataset to 4,920 news articles. Next, we preprocessed the news content by removing emojis, URL links, punctuations, numbers, and special tokens (e.g., "&"). We also converted the terms into single words and lowercase. Further, we removed standard stopwords from the NLTK library. Finally, we conducted a stemming process to enhance interpretability. After preprocessing, 43,792 valid English terms were prepared for further analyses described next.

3.2 Exploratory Data Analysis (EDA)

EDA is a fundamental step in data analysis that involves summarizing the main characteristics of a dataset. The main objective of EDA is to better understand the underlying data structure, identify patterns and relationships, and uncover hidden insights [31]. EDA uses various statistical and visualization techniques to explore the data, such as histograms, scatterplots, and word clouds. By performing EDA, researchers can make more informed decisions about further analyses, develop hypotheses, and refine models. In this study, we used histograms, frequency counting, and word clouds to explore the meaning of the data and inform our further analyses.

3.3 Topic Modelling Analysis

Latent Dirichlet allocation (LDA) is a popular and widely used machine learning method for text analysis. It is an unsupervised clustering technique to identify latent topics from a corpus and then calculate topic distributions [32]. In our study, we map all the documents to the classified topics and calculate the representatives of each topic. To make each document and each word as monochromatic as possible, we used the Gibbs sampling algorithm to increase the statistical inference [32]. Next, we apply Cao et al.'s [33] and Deveaud et al.'s [34] metrics to identify the optimal number of topics. Since the best topics number shows low values for Cao et al. [33] and high values for Deveaud et al. [34], we find the point that best fits the criteria. According to the metrics, we generated 11 topics in this study.

3.4 Sentiment Analysis

To address RQ2, we conducted sentiment analysis. Sentiment analysis is a broad term encompassing various approaches that extract emotion or opinion-related information from natural language [35]. This analytical technique has been applied in multiple disciplines, including psychology, economics, education, and political and social science [36]. Sentiment analysis provides information about positive, negative, or neutral attitudes, allowing us to investigate the author's attitude toward the topics. To obtain the sentiment of each news article, we first applied the VADER (Valence Aware Dictionary for Sentiment Reasoning) model to generate sentiment scores for each document [37]. Next, we categorized each document into three classifications according to the sentiment scores ($-0.5 < \text{neutral} < 0.5$, $\text{negative} \leq -0.5$, $\text{positive} \geq 0.5$).

4 Findings

Our findings are presented from three aspects. Section 4.1 provided an overview of our dataset from 2000 to 2022, including the distribution of publication sources and geographic information. Section 4.2 delves into a detailed description of the 11 topics generated through LDA analysis. Moreover, we demonstrate how these topics have evolved and diversified over the years, offering valuable insights into the changing landscape of deepfake news. Section 4.3 shifts our focus to illustrating sentiment distribution across both years and topics.

4.1 EDA

The emergence of deepfake news articles can be traced back to 2000, although it remained relatively scarce from 2000 to 2017, with an average of only two news articles per year. However, a significant shift occurred in 2018, marking a turning point in the landscape of deepfake news, as shown in Fig. 1. In 2018 and 2019, the number of news articles experienced a dramatic surge, reaching an unprecedented peak of 1,217. This surge can be attributed to various factors, including advancements in technology, increased accessibility to deepfake creation tools, and growing concerns surrounding the potential

consequences of this technology [2, 38, 39]. The years 2020 and 2021 witnessed a slight decline in deepfake news articles, potentially influenced by the COVID-19 pandemic and its impact on media coverage. Nevertheless, deepfake-related content regained momentum and reached a new pinnacle in 2022, with 1,486 news articles. These fluctuations in deepfake news reflect the development of deepfake and the public's increasing awareness of its implications.

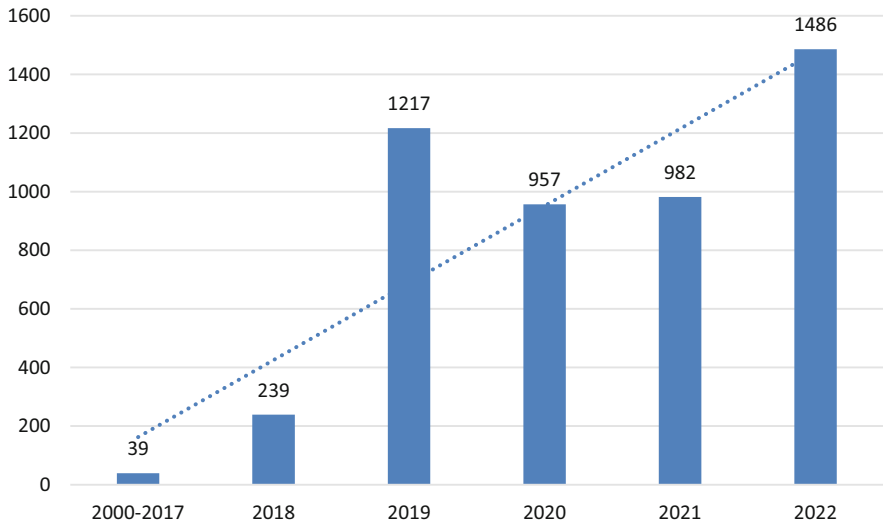


Fig. 1. News publishing trend over years

Over 500 news sources have published articles related to deepfake technology during 2000–2022. Table 1 shows the list of the top 20 sources and how these news publishers are distributed. Here, CE Noticias Financieras emerged as the most prolific publisher, account for 14.6% (720 news articles) of the total coverage. CE Noticias Financieras is renowned for its extensive daily coverage of Latin America, including Central, and South America, Mexico, with over 9,000 news stories published daily. The New York Times and The Independent have made similar contributions to deepfake-related news, accounting for approximately 4.3% and 4.2% of the articles, respectively. Notably, 12 are based in the United Kingdom among the top 20 sources. This concentration of sources from the UK could be attributed to the country's robust justice system [40].

We utilized word clouds to depict the significance of keywords based on frequency analysis visually. In the word cloud, the font size represents the frequency of each keyword's occurrence in the news content. Figure 2 presents the top 50 keywords ranked by their frequency. The keyword “say” holds the highest rank with a frequency of 24,205. Additionally, other highly recurring keywords include “video” (15,167), “use” (14,336), “make” (11,556), and “people” (10,598). These keywords unequivocally suggest that the primary application of deepfake technology lies in video production.

Table 1. Top 20 news sources

News source	No. of news
CE Noticias Financieras English	720
The New York Times	213
The Independent (United Kingdom)	207
The Guardian (London)	181
MailOnline	174
The Times (London)	140
Financial Times (London)	97
The Daily Telegraph (London)	77
thesun.co.uk	77
Hindustan Times	69
The Straits Times (Singapore)	63
The Sun (England)	63
thetimes.co.uk	58
Indian Express	56
Postmedia Breaking News	50
The Sunday Times (London)	46
National Post (f/k/a The Financial Post) (Canada)	46
The Times of India	43
mirror.co.uk	39
i-Independent Print Ltd	38

In the same way, we visually represent the geographic information that appeared most frequently in the news content. Figure 3 illustrates the frequencies of places mentioned in the news articles. The United States (USA) had the highest occurrence, with 2,648 mentions, followed by the United Kingdom (UK), with 1,263 mentions. China (541), India (463), Australia (371), and Canada (348) were also frequently referenced countries. It can thus be seen that news content related to deepfakes is mainly concentrated in the United States.



Fig. 2. Top 50 most frequently occurring terms



Fig. 3. Top 50 Geographic information mentioned in news content

4.2 Major Topics and Trends

As shown in Table 2, topics 1, 2, 3, 9, and 11 showcases the diverse applications of deepfake technology in areas such as marketing, politics, entertainment, military, and digital life. The utilization of deepfake technology has had a positive impact, opening up numerous possibilities and opportunities within society (Topic 5). However, it also raises concerns, including issues related to victimization and crime (Topic 6), the spread of mis/disinformation on social media (Topic 7), and cyber security (Topic 8). Despite being a double-edged sword, researchers and companies are dedicated to improving this technology while exploring ethical safeguards (Topic 10).

Table 2. 11 topics generated by topic modeling analysis

Topic label	Top 10 words	Rate	No. of Doc	Description
Topic 1: Marketing	company, user, app, channel, sell, message, service, tech, launch, brand	9.90%	287(5.83%)	Companies use deepfake technology to create immersive user experience
Topic 2: Political campaign	political, election, campaign, government, trump, candidate, party, vote, democracy, law	9.10%	398(8.09%)	Deepfake videos and audios persuade voters and threat to election
Topic 3: Entertainment usage	film, make, character, play, music, artist, fan, story, live, song	9.80%	500(10.16%)	Deepfakes in film and entertainment are common. It includes realistic-looking videos and images
Topic 4: Deepfake voice	video, face, voice, speak, actor, real, person, audio, clip, footage	8.00%	633(12.87%)	Deepfake voice refers to synthetic voice that closely mimics a real person's voice
Topic 5: Positive influence	world, good, work, change, society, student, future, idea, power, understand	8.70%	316(6.42%)	Deepfakes can create numerous possibilities and opportunities in society. For example, it increase efficiency and accessibility of work
Topic 6: Victim and crime	image, woman, online, victim, child, photo, police, abuse, crime, report	8.40%	475(9.65%)	A lot of women are victims of deepfake porn
Topic 7: Social media information	medium, social, news, content, facebook, twitter, disinformation, misinformation, spread,	9.20%	597(12.13%)	Deepfake are used to spread mid/disinformation on social media platforms
Topic 8: Cyber security	security, datum, threat, attack, cyber, information, risk, business, digital, increase	9.20%	460(9.35%)	Deepfake technology threat to cybersecurity. It can be applied to biometric authentication

(continued)

Table 2. (continued)

Topic label	Top 10 words	Rate	No. of Doc	Description
Topic 9: War and military	russian, country, war, force, claim, military, block, state, ukrainian, report	10.70%	322(6.54%)	Deepfake video used for propaganda and disinformation during Russia and Ukraine conflict
Topic 10: Technology development	human, image, create, system, generate, tool, develop, word, model, researcher	7.30%	377(7.66%)	Researchers endeavoured to develop research model and tools to fight deepfakes
Topic 11: digital life	life, series, good, deep, watch, leave, give, capture, family, episode	9.60%	555(11.28%)	Deepfake technology is used to resurrect deceased people

The line graph below demonstrates how deepfake topics have evolved and diversified over the years, and how specific topics gained more attention than others (see Fig. 4). For example, the sixth topic about victims and crime was the third least popular topic in 2019 but rapidly gained attention and rose to second place in 2022. Furthermore, Topic 7, which focuses on social media information, emerged as the most prominent topic in 2020, probably influenced by the pandemic and the widespread dissemination of mis/disinformation on social media platforms. Nevertheless, this topic experienced a significant decline and dropped to the fifth rank in 2022.

4.3 Sentiment Distribution

The results showed that the majority of deepfake news articles are positive in sentiment (50%), followed by negative (43%) and neutral (7%). The sentiment trends and distributions across year and topic were also examined, as shown in Fig. 5 and Fig. 6. In Fig. 5a, the distribution of news across three sentiments follows a similar trend as the total amount of news from 2000 to 2022. However, it is challenging to discern sentiment fluctuations over the years from this data alone. To address this, we calculated the percentage of sentiments for each month by dividing the number of articles in each sentiment category by the total number of news in that month. Our analysis revealed a slight rise in negative and neutral sentiments, accompanied by a corresponding decrease in positive sentiment over the years (see Fig. 5b).

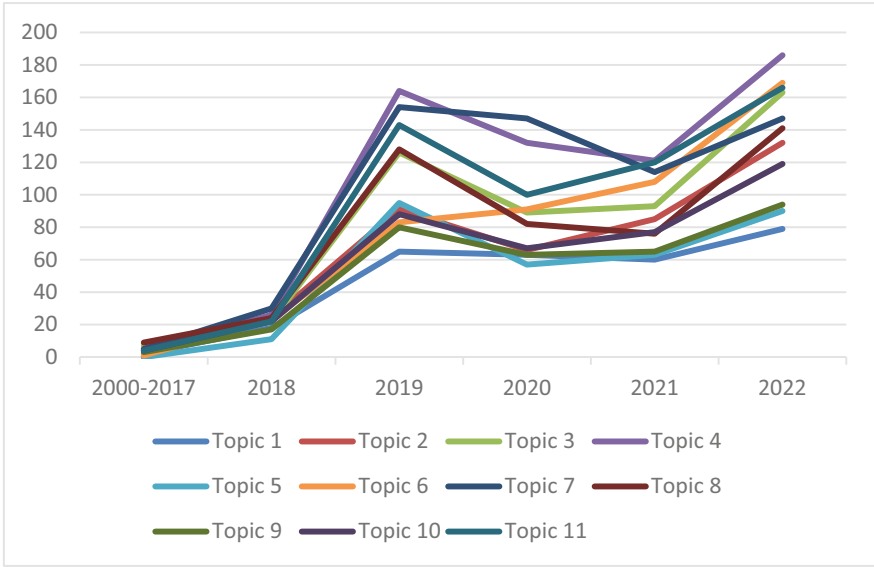
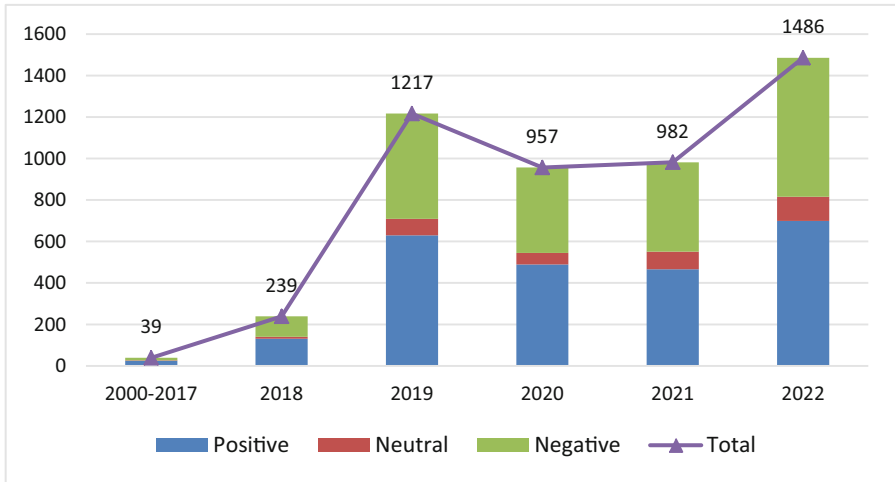
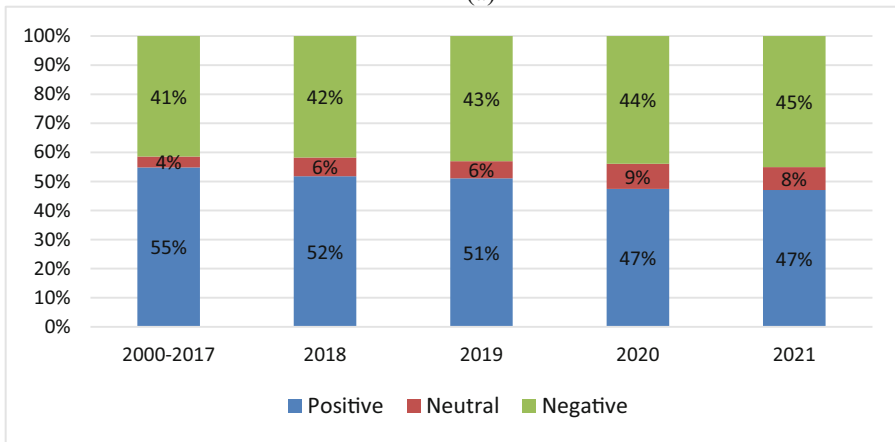


Fig. 4. Topic distribution over years

This finding suggests that deepfake-related content tends to exhibit an inclination towards negative or neutral sentiments, highlighting the potential concerns and challenges associated with this technology. Figure 6 reveals significant variations in sentiment distributions across different topics. Figure 6a demonstrates the amount of news in each sentiment category across topics. Our findings revealed that the top three positive news observed in Topic 3, 4 and 11, while the top three negative news observed in Topic 4, 6 and 7. To gain further insights, we calculated the percentage of sentiments in each topic by dividing the number of news pertaining to each topic by the total number of articles in that topic (see Fig. 6b). In Topic 6, Topic 7, and Topic 9, negative sentiment prevails over positive sentiment. On the other hand, in Topic 2, Topic 4, and Topic 8, the weightage of negative and positive sentiments is relatively balanced. In contrast, positive sentiment dominates in Topic 1, Topic 3, Topic 5, Topic 10, and Topic 11. These differences in sentiment distribution across topics highlight the diverse perspectives and attitudes associated with deepfake technology.

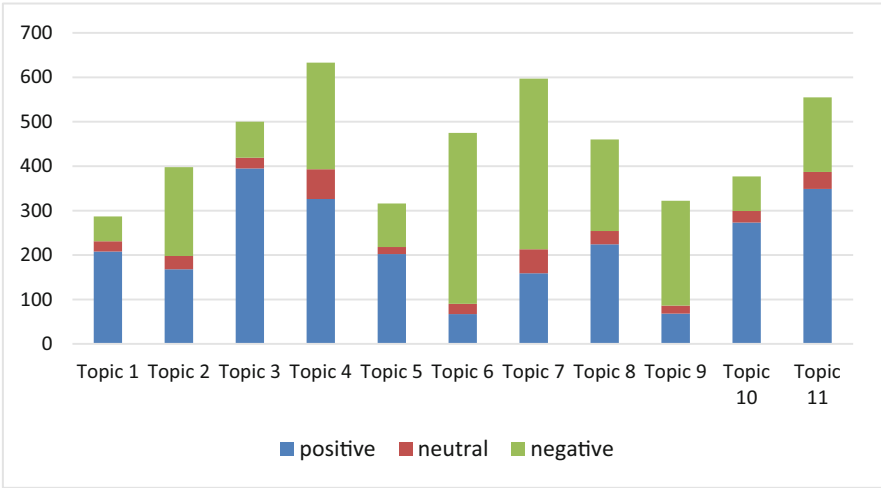


(a)

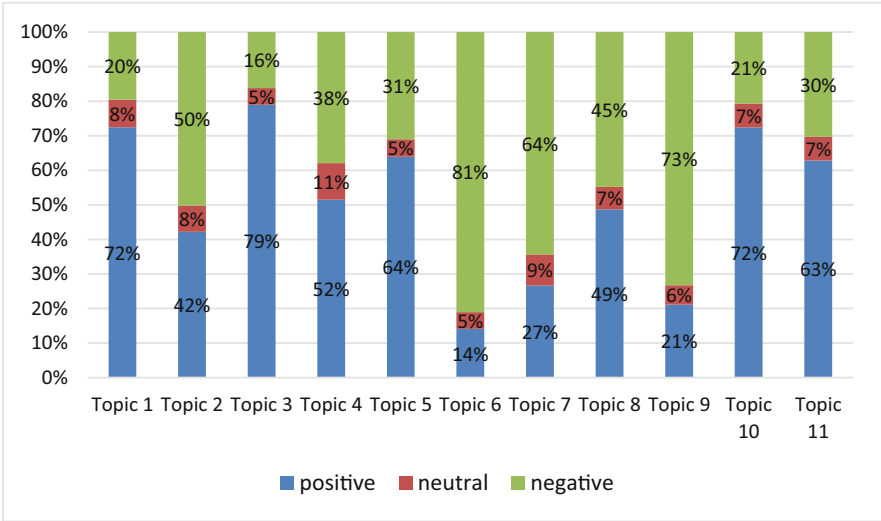


(b)

Fig. 5. Sentiment distribution by (a) No. of news (b) % of news



(a)



(b)

Fig. 6. Sentiment distribution by (a) No. of news (b) % of news in each topic

5 Discussion

The major topics identified through the topic modeling analysis encompass applications and concerns associated with deepfake technology. This study found that deepfake technology has been applied in different domains, as described in the news articles. Topics such as marketing, military, victimization and crime, cyber security, and digital life explore the wide-ranging uses of deepfakes. Deepfake technology can be leveraged in

marketing to create compelling and persuasive advertisements. The military topic suggests the potential of deepfakes for training simulations and strategic operations. While deepfake technology offers opportunities, there are concerns and ethical considerations. Topics such as victimization and crime draw attention to the possibility of using deepfakes for harmful purposes, including revenge porn, harassment, and extortion. The issue of cyber security also arises, as deepfakes can be utilized as a tool for cyber-attacks and identity theft.

Another finding is that deepfake topics change in focus over the years. For instance, victimization and crime gained prominence in 2022, indicating an increasing awareness of the potential harm caused by deepfakes. As deepfake technology becomes more sophisticated and accessible, there is a heightened risk of individuals being victimized by creating and disseminating manipulated videos [41]. Another significant shift in deepfake topics occurred in 2020, with a notable increase in the focus on social media misinformation. Deepfakes, with their ability to create realistic but fabricated videos, have the potential to amplify and spread false information at an alarming rate [42]. The rise of this topic in 2020 indicates the growing recognition of deepfakes' role in fueling misinformation campaigns and the urgent need for effective strategies to detect and combat them.

Moreover, our analysis of the distribution of publication sources provide insights into the extensive coverage of deepfake technology across the globe. With over 500 news sources publishing articles related to deepfakes, it is evident that this phenomenon has captured the attention of media outlets worldwide. Also, our geographic information analysis highlights the countries most frequently mentioned in deepfake news articles. The United Kingdom and the United States are the most referenced countries, and China, India, Australia, and Canada also receive notable mentions. These findings highlight the global reach and impact of deepfake news because its use and misuse have extended beyond national borders and resonated with media outlets globally. Our findings underscore the interconnection of deepfakes and emphasizes the need for international collaboration to develop strategies to mitigate the potential harms and promote the responsible use of this technology [43].

Furthermore, previous research has shown that deepfake technology has positively impacted society and daily life. It has been utilized in various areas, such as marketing campaigns, entertainment content, and increasing work efficiency [19, 27]. These benefits align with the findings of our study, which reveal a prevailing positive attitude towards this emerging technology in a large portion of news articles. In addition, much news content contained expectations for deepfake applications to enrich our society and enhance various aspects of our lives. For instance, an article published by The Sunday Herald sheds light on the capabilities of deepfake software and its potential for creating highly realistic face swaps in both images and videos, thereby driving advancements in the entertainment field.

In addition to the aforementioned advantages, our study revealed that the topic of digital life emerged prominently in recent deepfake news content. Many individuals have welcomed the integration of this technology and its potential applications. For instance, CE Noticias Financieras English reviewed deepfake application in avatar creation. The articles show that deepfakes have been used to generate realistic digital avatars that can

be employed in virtual reality environments, allowing users to better immerse themselves and interact with others more realistically online.

While there are generally positive attitudes towards deepfake technologies in Topics 1, 3, 5, 10, and 11, concerns have been growing regarding their potential for malicious use. These concerns include using deepfakes for political sabotage, market manipulation, and disseminating misinformation [2, 3]. Our findings in Topics 6, 7, and 9 also reflected similar concerns. For instance, an article released on April 22, 2019, in *The New York Times* expressed concerns about the spread of misinformation, offensive content, and disturbing videos as some of the challenges faced by YouTube. It also discussed controversies surrounding the platform, such as running ads alongside anti-vaccine content, enabling child suicide, and hosting videos with inappropriate content. These examples illustrate the potential negative consequences associated with the misuse of deepfake. It is thus crucial for platforms and policymakers to develop robust strategies for addressing these concerns and ensuring responsible use of the technology.

Lastly, the war between Ukraine and Russia has sparked heated discussions regarding deepfake technology in warfare and military operations. Undoubtedly, people have expressed negative opinions on its potential risks in such a context. The fake video or audio recordings could be employed to manipulate public opinion, deceive military forces, and spread disinformation during the conflict. One of the examples in our dataset stated, “Would Facebook allow distribution of a deepfaked video if it were shared by the president as the rationale for the bombing of Iran? What if the president shared a less manipulated “shallow fake” video like the Pelosi video of a potential 2020 opponent?” This example raises important questions about the extent to which platforms should permit the dissemination of manipulated media, particularly in high-stakes situations such as international conflicts and elections. The potential consequences and ethical implications of allowing such content to circulate freely necessitate careful consideration and robust policies to protect the public interest and prevent harmful outcomes.

6 Conclusion

This study analyzed news articles to gain insights into major topics and attitudes toward deepfakes. We aimed to understand the prevalence, frequency, and diverse range of deepfake-related issues discussed in news articles and the public’s perceptions and attitudes toward this emerging technology through the analysis. Our findings have implications for both theoretical understanding and practical applications. From a theoretical perspective, this study contributes to the existing literature by offering insights into the evolving landscape of deepfake-related news coverage. The significant increase in the number of deepfake news articles in recent years indicates a growing recognition of the consequences and risks of deepfakes. The findings align with previous research on the prevalence and impact of deepfakes in society.

The study’s topic modeling analysis uncovers the main themes and concerns discussed in news articles, providing a comprehensive overview of deepfake technology’s implications. This contributes to a nuanced understanding of its societal impacts. Additionally, our sentiment analysis offers insights into the emotional tone and attitudes conveyed in news articles, aiding in understanding public opinion as well as informing

future research on the social and psychological effects of deepfake technology. From a practical standpoint, the findings have implications for various stakeholders. Policymakers can benefit from the insights provided by our analysis of news articles, facilitating the development of appropriate regulations, guidelines, and awareness campaigns to mitigate the negative impacts of deepfakes. Media organizations and journalists can utilize our analysis to understand prevailing topics, narratives, and attitudes surrounding deepfakes. This understanding enables the development of informed and responsible reporting practices, ensuring accurate and balanced coverage of deepfake-related issues, thereby fostering media literacy among the general public.

However, our study has some limitations. The analysis was based solely on news articles, which may not capture the full range of public opinion and attitudes toward deepfakes. Other sources, such as social media or public forums, could provide additional insights. Also, the analysis focused on English-language news articles, which may limit the generalizability of the findings to other languages and regions. Future work in this area could expand the study to include a broader range of sources, such as online forums, to capture a more comprehensive understanding of public perceptions and attitudes toward deepfakes. Additionally, conducting comparative studies across different languages and cultures would provide insight into the cultural variations in attitudes toward deepfakes.

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